



FLOOD RISK ASSESSMENT



2409-955 Sladburys Lane, Holland-on-sea

Flood Risk Assessment
& Drainage Strategy

TABLE OF CONTENTS

Table of Contents	1
Document Control Sheet	2
1. Introduction	3
Project	3
Aims and objectives	3
Site location	3
2. Flood analysis	4
Flood level.....	4
Flood protection.....	4
Extreme flood events	6
3. Surface water drainage	7
Drainage and Site Analysis.....	7
Surface water strategy	7
4. Conclusions and recommendations.....	8
5. SuDS Design and Management	9
Water Quality, Amenity and Biodiversity.....	9
Maintenance of SuDS	11
SuDS Mitigation	15
Management of surface water during construction	16
8. Appendix A – Drainage strategy	17
9. Appendix B – Topographical survey	18
10. Appendix C – Sewer records.....	19
11. Appendix D –Hydraulic calculations	20
12. Appendix E –Geotechnical Report	21

DOCUMENT CONTROL SHEET

REV	ISSUE PURPOSE	AUTHOR	CHECKED	APPROVED	DATE
-	Initial submission	JL	KK	KK	18.06.25

1. INTRODUCTION

Project

Ingent Consultant Engineers have been appointed on behalf of Stanfords for the application for 29 new residential dwellings to be constructed on a previous greenfield site located off Sladburys Lane, Holland-On-Sea. The scheme currently does not have planning permission, and this report is submitted in support of the proposed outline planning application submission. The application is for the development of up to 29 dwellings, its means of access off Sladburys Lane, open space and associated infrastructure.

The development is to comprise of a mixture of detached, semi-detached and flats, and the proposal will include the new housing, parking, hardstanding's access via a proposed bell mouth connection of Sladburys Lane, onto a site area of approximately 1.56Ha. (Please refer to the Appendix for the proposed layout shown on the drainage strategy 2409-955-ST001).

Aims and objectives

The purpose of this assessment is to review the proposed development and to consider the risk to it of flooding, also the affect it may have on existing flood flows and flood storage within adjacent floodplains.

The risks of flooding and potential impact of alterations to the existing flood flow and storage regime and surface run-off characteristics are considered within the scope and parameters set out in the National Planning Policy Framework (NPPF). Consideration is also given to the ODPM document "Preparing for Floods".

The CIRIA C753 SuDS manual, Essex LLFA - Sustainable Drainage Systems (SuDS) a Local Design Guide and Environment Agency websites flood mapping was consulted during this assessment. (Please refer to the 'Flood Analysis' section of this report for flood mapping data).

Site location

The site is located in Holland-on-sea, which is a town in the county of Essex, on the east coast of England. It is 82 miles (132 km) north east of central London. The proposed development is approximately 1.7 miles east of the A133 and borders fields to the north boundary and a new development bordering the west boundary. The site has an existing ditch on the north, east and partly south of the site and an existing river 170 meters south of the site.

The site falls from 10.52m along the western boundary to 5.6m in the east, gradients are fairly consistent in the region of 1in35 (2.8%) falling in an easterly direction. (Please refer to the Appendix detailing the Topographical Survey on the drainage strategy drawing)

2. FLOOD ANALYSIS

Flood level

The Environment Agency flood mapping (as shown below) indicates that the site lies Flood Zone 1, outside any floodplain with <1in1000yr probability of river or sea flooding.



Flood protection

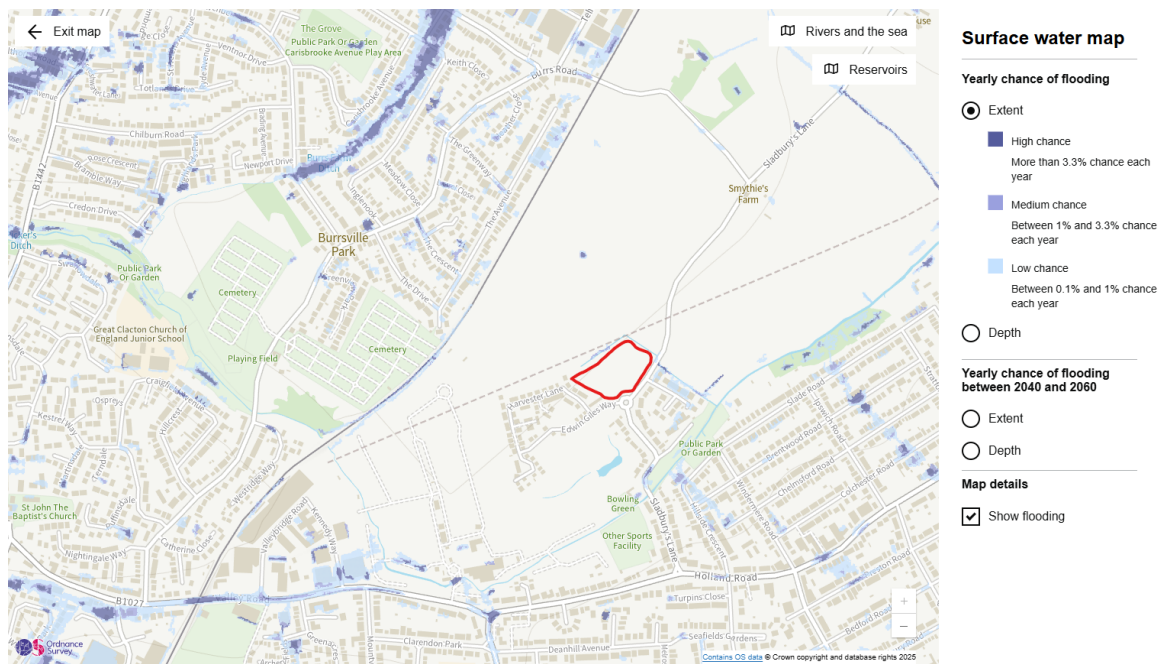
Although within flood zone 1, consideration is also required of the flood risk presented from other potential sources such as overland water, groundwater, sewers and retained water features. The site, proposed levels and existing topography of the site have been assessed for these threats, such that in the event of large storm events, surface water flows would not threaten life or property. A detailed list of the flood risks considered is as follows:

Overland water: There is an existing river which does not represent a risk to this site. The nearest river system is approximately 170m to the south however flood mapping shows the site does not lie within the flood risk area. An existing ditch borders the site however flood mapping shows no risk of flooding and that the ditch captures the surface water run-off.

Groundwater: Where the water table rises to such a height where flooding occurs. Most common in low-lying areas underlain by permeable ground (aquifers), usually due to extended periods of wet weather: Ground Water flooding is not considered to have any impact as the borehole records show that groundwater was not recorded during drilling up to depths of approximately 10 meters below existing ground level. The geology mapping sourced also confirmed the site to lie on clay, meaning the site will not infiltrate, therefore the use of a detention basin and offsite discharge is required. (see Appendix).

Sewers and highway drains: Combined, foul or surface water sewers and highway drains that are temporarily over-loaded due to excessive rainfall or due to blockage: Should the proposed onsite sewers become blocked, the proposed road levels and garden levels surrounding the buildings will be set lower than the proposed finished floor levels therefore ensuring that there is no threat to adjacent property.

Surface water: The net rainfall falling on a surface (on or off the site) which acts as runoff which has not infiltrated into the ground or entered into a drainage system: As shown on the Flood Map for Planning the site has no risk of surface water building up on the site before flowing east to the sites low point and discharging into the existing ditch.



The appended drainage strategy details how all surface water run-off will drain via adoptable surface water drainage to a detention basin on site with storage for up to 1in100year + 45% climate change being provided within an allowance of 10% is added to roof areas for consideration of urban creep.

Infrastructure failure: canals, reservoirs, industrial processes, burst water mains or failed pumping stations: No infrastructure present to pose a threat.

As the site is above the 1% flood level, the proposed development is not considered to have any impact on existing floodplain storage.

Extreme flood events

Consideration is given to flood events over and above the 1in100yr event and exceedance flow routes. All flows will be directed away from properties with finished floor levels set to a minimum of 150mm above ground level, low spots will be designed in the hard standing directed away from buildings, ponding will give prior warning if failure of the drainage system occurs.

Flood mapping shows both the site and surrounding areas do not experience any surface water flooding. However, the ditch networks around the site boundary would intercept any potential for overland flows protecting the site in any extreme event.

The drainage strategy accommodates all surface water run-off up to 1in100year storm plus 45% climate change which will be contained in the detention basin whilst it discharges into the existing ditch, with an allowance of 10% added to roof areas for consideration of urban creep.

Although not used in design calculations there will be additional storage volume in the piped network and freeboard in the detention basin which can accommodate further run-off above the 1in100year storm plus 45% climate change plus 10% urban creep.

3. SURFACE WATER DRAINAGE

Drainage and Site Analysis

The borehole testing and geology mapping sourced from British Geological Surveying details no ground water encountered, up to depths of 10m and the underlying geology to be clay, therefore as clays soils tend to exhibit poor drainage characteristics infiltration is not considered to be viable for the site as a method of discharge for surface water.

Considering the above and following the hierarchy set out in NPPF the surface water strategy will be to discharge to the ditch network mimicking the existing drainage regime of the site.

The NPPF states the hierarchy for discharge of surface water to be: -

1. Into the ground (infiltration)
2. To the surface water body
3. To the surface water sewer, highway drain, or another drainage system
4. To a combined sewer

From the site analysis, there is presently 0m² of impermeable area, which post development will increase to approximately 6034m² with the existing drainage regime at present that surface water run-off discharges to the existing ditch via gravity.

Surface water strategy

The proposed surface water strategy is to convey flows via private pipe network, with permeable paving for large, shared parking areas, swales for the main adopted entrance road connecting to an adoptable gravity surface water sewers which discharges to a new detention basin constructed in the south-east low-lying area of the site, with eventual outfall to the sites southern ditch.

The basin will include sediment forebay, low flow channels and reed beds to provide additional treatment, the basin is designed to accommodate attenuation up to and including a 1 in 100-year storm event plus a 45% allowance for climate change and consideration of urban creep (10% for roof areas). Flows from the site will be restricted to the 1in1 year greenfield runoff rate of 1.8l/s based on the total site area, these features will not only reduce demand on the existing watercourse but provide additional storage and reduce the flow rate of surface run-off from the development.

The strategy is therefore considered to provide a robust solution to surface water drainage for the proposed development with all water contained in such a way as to pose no threat to life or property. The proposed system can accommodate surface water from events up to and including the extreme 1 in 100-year event plus a 45% allowance for climate change and consideration for urban creep as per appended example calculations.

The appended drainage strategy drawing 2409-955-ST001 (DRAINAGE STRATEGY) illustrates how the required attenuation may be located within the development.

4. CONCLUSIONS AND RECOMMENDATIONS

The proposed new residential development has been assessed for the risk to the site from fluvial flooding in accordance with the guidelines of the NPPF. Assessment has also been made of the impact of the proposed development on the existing floodplain storage at this location, and of the risk of pluvial flooding from storm water contribution from the development exacerbating downstream flooding, as directed by the guidance.

The site has been found to lie within Flood Zone 1 and is therefore not considered to be at risk from fluvial flooding. Other sources of flooding have similarly been assessed and found to pose no threat to development on the site.

The proposals do not affect flood storage within the floodplain and the peak surface water runoff will be reduce to the 1in1 year rate via storage in attenuation devices, which therefore reduces the demand on the existing watercourse network.

It is considered from this assessment that the discharge of future surface water runoff from the site will not be increased because of the redevelopment but will provide a betterment in peak flows discharging from site, by attenuating and slowing the rate at which it discharges into the existing watercourse, and the level of risk of flooding to and from the proposed development is of an acceptable level and in keeping with current policy.

The proposed drainage strategy is therefore considered to provide a robust solution to surface water drainage for the proposed development with all surface water contained in such a way as to pose no threat to life or property. The proposed system can accommodate surface water run-off from all events up to and including the extreme 1 in 100-year event plus a 45% allowance for climate change, as per appended example calculations.

5. SUDS DESIGN AND MANAGEMENT

Water Quality, Amenity and Biodiversity

The approach to Sustainable Drainage Systems (SuDS) is to control water quantity, water quality, amenity and biodiversity by incorporating multiple SuDS features into a drainage design and producing a management train it is possible to achieve enhancements in all areas.

All hard standing is to utilise trapped gullies and catch pit manholes to provide a pre-treatment stage for surface water run-off prior to discharging to the surface water network.

It is proposed for all roof water to drain to a private pipe network discharging to an adoptable network subsequently conveyed to a detention basin which will include, sediment forebays, low flow channels and reed beds.

These combined measures produce a management train, which facilitates the capture, conveyance, and storage of surface water run-off, while delivering interception and pollution risk management.

The use of SUDS is proposed within the surface water drainage strategy for the proposed development of 29 new residential dwellings, in order to serve several functions:

- Control of runoff water rates and volumes, thereby reducing downstream flood risk
- Reducing pollutant concentrations and act as a buffer for accidental spills to improve runoff water quality

The range of SUDs described are listed in the below table 'Types of SuDS Components' and are described further in the maintenance of SuDS section of this document.

TABLE Types of SuDS components

1.1

Component type	Description	Assessment for proposed development
Rainwater harvesting systems	Rainwater is collected from the roof of a building or from other paved surfaces in an over-ground or underground tank for use on site. Depending on its intended use, the system may include treatment elements. The system should include specific storage provision if it is to be used to manage runoff to a design standard.	Not required as store in permeable paving subbase
Green roofs	A planted soil layer is constructed on the roof of a building to create a living surface. Water is stored in the soil layer and absorbed by vegetation. Blue roofs store water at roof level, without the use of vegetation.	Not appropriate
Infiltration systems	These systems collect and store runoff allowing it to infiltrate into the ground. Overlying vegetation and underlying unsaturated soils can offer protection to groundwater from pollution risks.	Infiltration is not viable on this site
Proprietary treatment systems	These subsurface and surface structures are designed to provide treatment of water through the removal of contaminants.	Not required
Filter strips	Runoff from an impermeable area is allowed to flow across a grassed or otherwise densely planted area to promote sedimentation and filtration.	Incorporated into the development
Filter drains	Runoff is temporarily stored below the surface in a shallow trench filled with stone/gravel, providing attenuation, conveyance and treatment (via filtration).	Incorporated into the development beneath swale
Swales	A vegetated channel is used to convey and treat runoff (via filtration). These can be "wet", where water is designed to remain permanently at the base of the swale, or "dry" where water is only present in the channel after rainfall events. It can be lined, or unlined to allow infiltration.	Incorporated into the development
Bioretention systems	A shallow landscaped depression allows runoff to pond temporarily on the surface, before filtering through vegetation and underlying soils prior to collection or infiltration. In its simplest form it is often referred to as a rain garden. Engineered soils (gravel and sand layers) and enhanced vegetation can be used to improve treatment performance.	Incorporated into the development
Trees	Trees can be planted within a range of infiltration SuDS components to improve their performance, as root growth and decomposition increase soil infiltration capacity. Alternatively they can be used as standalone features within soil-filled tree pits, tree planters or structural soils, collecting and storing runoff and providing treatment (via filtration and phytoremediation).	Infiltration is not viable on this site
Pervious pavements	Runoff is allowed to soak through structural paving. This can be paving blocks with gaps between solid blocks, or porous paving where water filters through the block itself. Water can be stored in the sub-base and potentially allowed to infiltrate into the ground.	Incorporated into the development
Attenuation storage tanks	Large, below-ground voided spaces can be used to temporarily store runoff before infiltration, controlled release or use. The storage structure is often constructed using geocellular or other modular storage systems, concrete tanks or oversized pipes.	Not required as storage is within above ground SuDS or permeable paving subbase
Detention basins	During a rainfall event, runoff drains to a landscaped depression with an outlet that restricts flows, so that the basin fills and provides attenuation. Generally, basins are dry, except during and immediately following the rainfall event. If vegetated, runoff will be treated as it is conveyed and filtered across the base of the basin.	Incorporated into the site wide strategy
Ponds and wetlands	Features with a permanent pool of water can be used to provide both attenuation and treatment of runoff, where outflows are controlled and water levels are allowed to increase following rainfall. They can support emergent and submerged vegetation along their shoreline and in shallow, marshy zones, which enhances treatment processes and biodiversity.	Incorporated into the site wide strategy but limited due to available space

Maintenance of SuDS

Detention/Balancing basins are depressions in the ground that store water. Water levels rise after rain and then drop to the normal level as the excess is slowly released into the adjacent watercourse at a controlled rate. The detention basin is usually dry although in this case will have a low flow channel for the passage of water in non-storm events. Maintenance required is as follows:

TABLE 22.1 Operation and maintenance requirements for detention basins

Maintenance schedule	Required action	Typical frequency
Regular maintenance	Remove litter and debris	Monthly
	Cut grass – for spillways and access routes	Monthly (during growing season), or as required
	Cut grass – meadow grass in and around basin	Half yearly (spring – before nesting season, and autumn)
	Manage other vegetation and remove nuisance plants	Monthly (at start, then as required)
	Inspect inlets, outlets and overflows for blockages, and clear if required.	Monthly
	Inspect banksides, structures, pipework etc for evidence of physical damage	Monthly
	Inspect inlets and facility surface for silt accumulation. Establish appropriate silt removal frequencies.	Monthly (for first year), then annually or as required
	Check any penstocks and other mechanical devices	Annually
	Tidy all dead growth before start of growing season	Annually
	Remove sediment from inlets, outlet and forebay	Annually (or as required)
	Manage wetland plants in outlet pool – where provided	Annually (as set out in Chapter 23)
Occasional maintenance	Reseed areas of poor vegetation growth	As required
	Prune and trim any trees and remove cuttings	Every 2 years, or as required
	Remove sediment from inlets, outlets, forebay and main basin when required	Every 5 years, or as required (likely to be minimal requirements where effective upstream source control is provided)
Remedial actions	Repair erosion or other damage by reseedling or re-turfing	As required
	Realignment of rip-rap	As required
	Repair/rehabilitation of inlets, outlets and overflows	As required
	Relevel uneven surfaces and reinstate design levels	As required

Swales – Are shallow, flat bottomed, vegetated and open channels designed to convey, treat and often attenuate surface water runoff.

Maintenance schedule	Required action	Typical frequency
Regular maintenance	Remove litter and debris	Monthly, or as required
	Cut grass – to retain grass height within specified design range	Monthly (during growing season), or as required
	Manage other vegetation and remove nuisance plants	Monthly at start, then as required
	Inspect inlets, outlets and overflows for blockages, and clear if required	Monthly
	Inspect infiltration surfaces for ponding, compaction, silt accumulation, record areas where water is ponding for > 48 hours	Monthly, or when required
	Inspect vegetation coverage	Monthly for 6 months, quarterly for 2 years, then half yearly
	Inspect inlets and facility surface for silt accumulation, establish appropriate silt removal frequencies	Half yearly
Occasional maintenance	Reseed areas of poor vegetation growth, alter plant types to better suit conditions, if required	As required or if bare soil is exposed over 10% or more of the swale treatment area
Remedial actions	Repair erosion or other damage by re-turfing or reseedling	As required
	Relevel uneven surfaces and reinstate design levels	As required
	Scarify and spike topsoil layer to improve infiltration performance, break up silt deposits and prevent compaction of the soil surface	As required
	Remove build-up of sediment on upstream gravel trench, flow spreader or at top of filter strip	As required
	Remove and dispose of oils or petrol residues using safe standard practices	As required

Pervious Pavements

Pervious pavements including permeable block paving, porous asphalt or gravel and allow rain to percolate through the surface into underlying drainage layers. They should be protected from silt, sand, compost, mulch etc. Permeable block paving and porous asphalt can be cleaned by suction brushing

Maintenance schedule	Required action	Typical frequency
Regular maintenance	Brushing and vacuuming (standard cosmetic sweep over whole surface)	Once a year, after autumn leaf fall, or reduced frequency as required, based on site-specific observations of clogging or manufacturer's recommendations – pay particular attention to areas where water runs onto pervious surface from adjacent impermeable areas as this area is most likely to collect the most sediment
Occasional maintenance	Stabilise and mow contributing and adjacent areas	As required
	Removal of weeds or management using glyphosate applied directly into the weeds by an applicator rather than spraying	As required – once per year on less frequently used pavements
Remedial Actions	Remediate any landscaping which, through vegetation maintenance or soil slip, has been raised to within 50 mm of the level of the paving	As required
	Remedial work to any depressions, rutting and cracked or broken blocks considered detrimental to the structural performance or a hazard to users, and replace lost jointing material	As required
	Rehabilitation of surface and upper substructure by remedial sweeping	Every 10 to 15 years or as required (if infiltration performance is reduced due to significant clogging)
Monitoring	Initial inspection	Monthly for three months after installation
	Inspect for evidence of poor operation and/or weed growth – if required, take remedial action	Three-monthly, 48 h after large storms in first six months
	Inspect silt accumulation rates and establish appropriate brushing frequencies	Annually
	Monitor inspection chambers	Annually

Bioretention Systems

Bioretention systems (including rain gardens) are shallow landscaped depressions that can reduce runoff rates and volumes and treat pollution through the use of engineered soils and vegetation. They are particularly effective in delivering Interception and can also provide:

- attractive landscape features that are self-irrigating and fertilising
- habitat and biodiversity
- cooling of the local microclimate due to evapotranspiration.

Maintenance schedule	Required action	Typical frequency
Regular inspections	Inspect infiltration surfaces for silting and ponding, record de-watering time of the facility and assess standing water levels in underdrain (if appropriate) to determine if maintenance is necessary	Quarterly
	Check operation of underdrains by inspection of flows after rain	Annually
	Assess plants for disease infection, poor growth, invasive species etc and replace as necessary	Quarterly
	Inspect inlets and outlets for blockage	Quarterly
Regular maintenance	Remove litter and surface debris and weeds	Quarterly (or more frequently for tidiness or aesthetic reasons)
	Replace any plants, to maintain planting density	As required
	Remove sediment, litter and debris build-up from around inlets or from forebays	Quarterly to biannually
Occasional maintenance	Infill any holes or scour in the filter medium, improve erosion protection if required	As required
	Repair minor accumulations of silt by raking away surface mulch, scarifying surface of medium and replacing mulch	As required
Remedial actions	Remove and replace filter medium and vegetation above	As required but likely to be > 20 years

SuDS Mitigation

To deliver adequate treatment, SuDS components should have a total pollution mitigation index for each contamination type that is greater than or equal to the pollution hazard index as detailed in the following tables from CIRIA SuDS Manual (C753) 2016

As the proposed development has car parking pollution hazard indices are taken from the below table from with the areas relevant highlighted.

TABLE 26.2	Pollution hazard indices for different land use classifications				
	Land use	Pollution hazard level	Total suspended solids (TSS)	Metals	Hydro-carbons
	Residential roofs	Very low	0.2	0.2	0.05
	Other roofs (typically commercial/ industrial roofs)	Low	0.3	0.2 (up to 0.8 where there is potential for metals to leach from the roof)	0.05
	Individual property driveways, residential car parks, low traffic roads (eg cul de sacs, homezones and general access roads) and non-residential car parking with infrequent change (eg schools, offices) ie < 300 traffic movements/day	Low	0.5	0.4	0.4
	Commercial yard and delivery areas, non-residential car parking with frequent change (eg hospitals, retail), all roads except low traffic roads and trunk roads/motorways ¹	Medium	0.7	0.6	0.7
	Sites with heavy pollution (eg haulage yards, lorry parks, highly frequented lorry approaches to industrial estates, waste sites), sites where chemicals and fuels (other than domestic fuel oil) are to be delivered, handled, stored, used or manufactured; industrial sites; trunk roads and motorways ¹	High	0.8 ²	0.8 ²	0.9 ²

These index figures are compared to the SuDS mitigation for discharge to surface water, as detailed earlier in this report the surface water strategy is to have permeable paving, therefore the SuDS mitigation is as highlighted below. A factor of 0.5 is used for subsequent features for the reduced performance of secondary components.

TABLE 26.3 Indicative SuDS mitigation indices for discharges to surface waters

Type of SuDS component	Mitigation indices ¹		
	TSS	Metals	Hydrocarbons
Filter strip	0.4	0.4	0.5
Filter drain	0.4 ²	0.4	0.4
Swale	0.5	0.6	0.6
Bioretention system	0.8	0.8	0.8
Permeable pavement	0.7	0.6	0.7
Detention basin	0.5	0.5	0.6
Pond ⁴	0.7 ³	0.7	0.5
Wetland	0.8 ³	0.8	0.8
Proprietary treatment systems ^{5,6}	These must demonstrate that they can address each of the contaminant types to acceptable levels for frequent events up to approximately the 1 in 1 year return period event, for inflow concentrations relevant to the contributing drainage area.		

As the Total SuDS mitigation index is greater the pollution hazard index for each component the SuDS mitigation is met inline with CIRIA SuDS Manual (C753) 2016

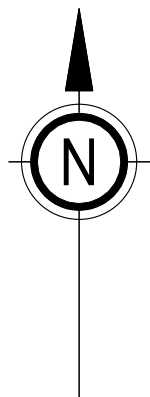
Management of surface water during construction

The requirement for management of surface water during construction will be highlighted to contractors at all stages of construction.

Specific issues to be addressed and the methods that might be used are all detailed in the following chapter **Construction Surface Water Management Plan**.

The following chapter should be used as guidance for contractors at tender stage and they will be expected to produce method statements responding to their Plan and include allowance within their pricing to implement the agreed method statements.

8. APPENDIX A – DRAINAGE STRATEGY



GENERAL

1. DO NOT SCALE THIS DRAWING. ALL DIMENSIONS ARE IN METRES UNLESS OTHERWISE NOTED. ANY DISCREPANCIES ARE TO BE RECORDED AND REPORTED TO THE ENGINEERS IMMEDIATELY.
2. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH ALL OTHER ENGINEERS AND ARCHITECTS DRAWINGS AND THE SPECIFICATION.
3. ANY EXISTING DETAILS WHICH ARE SHOWN ON THIS DRAWING ARE FOR GUIDANCE ONLY AND ARE TO BE CHECKED ON SITE BY THE CONTRACTOR. ANY VARIATIONS ARE TO BE RECORDED AND REPORTED TO THE ENGINEER IMMEDIATELY.

NOTES

1. AN ALLOWANCE OF 10% HAS BEEN ADDED TO IMPERMEABLE ROOF AREAS FOR THE CONSIDERATION OF URBAN CREEP.
2. THE SURFACE WATER NETWORK HAS BEEN DESIGNED TO ACCOMMODATE A 11N100YR STORM EVENT TO INCLUDE 45% CLIMATE CHANGE.
3. ALL ADOPTABLE SURFACE WATER SEWERS WITHIN ADOPTABLE HIGHWAYS ARE TO BE A MINIMUM OF 225MM.
4. ALL SEWER WORKS FOR THE WATER AUTHORITY ARE TO BE IN ACCORDANCE WITH 'SEWERAGE SECTOR GUIDANCE APPENDIX C' DESIGN AND CONSTRUCTION GUIDANCE VERSION 1.0 25TH OCTOBER 2019 AND ANY ADDENDUM THEREAFTER.
5. ALL PRIVATE CONNECTIONS TO ADOPTABLE SEWERS SHOULD BE A MINIMUM OF 100MM DIAMETER (FOUL) & 150MM DIAMETER (SURFACE) MADE IN VITRIFIED CLAY CONNECTING SOFFIT TO SOFFIT.
6. THIS DRAWING IS FOR INTENDED FOR STRATEGY PURPOSES ONLY TO ASSIST AND INFORM IN OUTLINING THE INITIAL CONCEPTS AND FEASIBILITY OF THE SCHEME, ANY DETAILS OR INFORMATION ON THIS DRAWING ARE SUBJECT TO CHANGE.

Rev	Revision Notes	Dwn	Apvd	Date

Unit 10 Brightwell Barns, Waldringfield Road, Brightwell, Suffolk, IP10 0BJ

STRATEGY

Project:
**LAND NORTH OF SLADBURY'S LANE
CLACTON-ON-SEA, ESSEX.**

Drawing Title:
SURFACE WATER DRAINAGE STRATEGY

Client: STANFORDS			Date: JUNE 26
Drawn: JL	Designed: KM	Checked: KK	Approved: KK

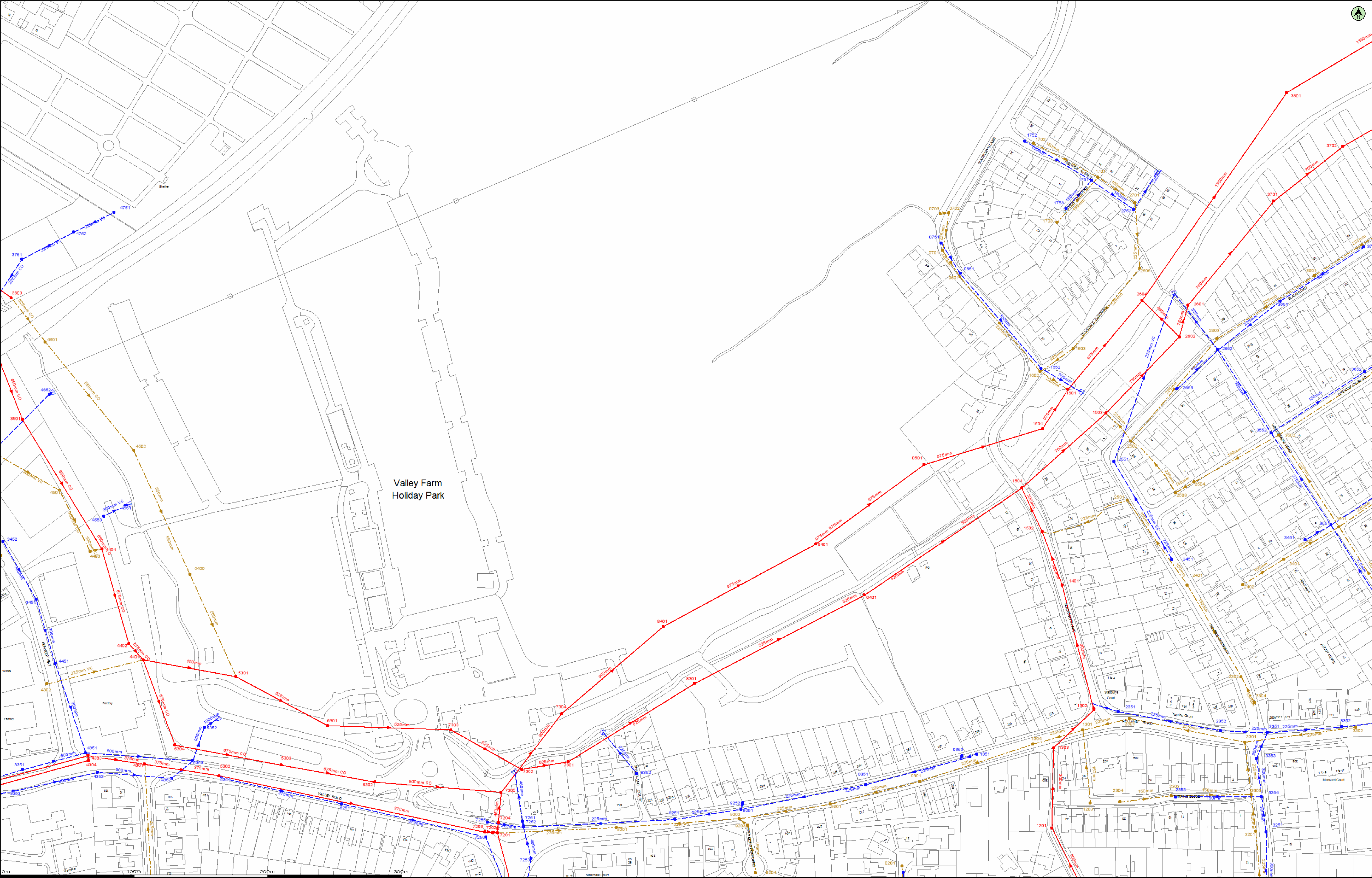
Scale: 1:500

Project No: 2409-955	Drawing No & Revision: ST001	Size: A2
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9. APPENDIX B – TOPOGRAPHICAL SURVEY



10. APPENDIX C – SEWER RECORDS



(c) Crown Copyright and database rights 2014 Ordnance Survey 100022432 Date: 04/02/15 Scale: 1:1250 Map Centre: 618884,216572 Data updated: 02/01/15 Our Ref: 128429 - 1 Wastewater Plan A1

This plan is provided by Anglian Water pursuant to its obligations under the Water Industry Act 1991 sections 198 or 199. It must be used in conjunction with any search results attached. The information on this plan is based on data currently recorded but position must be regarded as approximate. Service pipes, private sewers and drains are generally not shown. Users of this map are strongly advised to commission their own survey of the area shown on the plan before carrying out any works. The actual position of all apparatus MUST be established by trial holes. No liability whatsoever, including liability for negligence, is accepted by Anglian Water for any error or inaccuracy or omission, including the failure to accurately record, or record at all, the location of any water main, discharge pipe, sewer or disposal main or any item of apparatus. This information is valid for the date printed. The plan is produced by Anglian Water Services Limited from Ordnance Survey © Crown Copyright, 100022432. This map is to be used for the purposes of viewing the location of Anglian Water plant only. Any other uses of the map data or further copies is not permitted. This notice is not intended to exclude or restrict liability for death or personal injury resulting from negligence.

- | | | |
|----------------------|-----------|--------------------------------|
| Foul Sewer | — — — — — | Outfall |
| Surface Sewer | — — — — — | (Colour denotes effluent type) |
| Combined Sewer | — — — — — | Inlet |
| Private Sewer | — — — — — | (Colour denotes effluent type) |
| Decommissioned Sewer | — — — — — | Manhole |
| | — — — — — | (Colour denotes effluent type) |
| | — — — — — | Sewage Treatment Works |
| | — — — — — | Pumping Station |

●	Burr Road, Clacton
●	
●	



11. APPENDIX D –HYDRAULIC CALCULATIONS

Design Settings

Rainfall Methodology	FEH-22	Time of Entry (mins)	5.00	Connection Type	Level Soffits	Enforce best practice design rules	x
Return Period (years)	2	Maximum Time of Concentration (mins)	30.00	Minimum Backdrop Height (m)	9.900		
Additional Flow (%)	0	Maximum Rainfall (mm/hr)	50.0	Preferred Cover Depth (m)	0.900		
CV	1.000	Minimum Velocity (m/s)	1.00	Include Intermediate Ground	✓		

Nodes

Name	Area (ha)	T of E (mins)	Cover Level (m)	Diameter (mm)	Easting (m)	Northing (m)	Depth (m)
BASIN	0.622	5.00	6.000		619128.497	216856.960	1.200
HYDROBRAKE			6.000	1350	619133.792	216859.807	1.200
OUTFALL			5.600		619140.312	216856.785	0.855

Links

Name	US Node	DS Node	Length (m)	ks (mm) / n	US IL (m)	DS IL (m)	Fall (m)	Slope (1:X)	Dia (mm)	T of C (mins)	Rain (mm/hr)
1.000	BASIN	HYDROBRAKE	6.012	0.600	4.800	4.800	0.000	0.0	375	5.10	48.9
1.001	HYDROBRAKE	OUTFALL	7.186	0.600	4.800	4.745	0.055	130.7	150	5.24	48.5

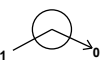
Name	Vel (m/s)	Cap (l/s)	Flow (l/s)	US Depth (m)	DS Depth (m)	Σ Area (ha)	Σ Add Inflow (l/s)	Pro Depth (mm)	Pro Velocity (m/s)
1.000	1.000	110.4	109.9	0.825	0.825	0.622	0.0	0	∞
1.001	0.877	15.5	109.0	1.050	0.705	0.622	0.0	150	0.894

Pipeline Schedule

Link	Length (m)	Slope (1:X)	Dia (mm)	Link Type	US CL (m)	US IL (m)	US Depth (m)	DS CL (m)	DS IL (m)	DS Depth (m)
1.000	6.012	0.0	375	DSG SW Pipe	6.000	4.800	0.825	6.000	4.800	0.825
1.001	7.186	130.7	150	DSG SW Pipe	6.000	4.800	1.050	5.600	4.745	0.705

Link	US Node	Dia (mm)	Node Type	MH Type	DS Node	Dia (mm)	Node Type	MH Type
1.000	BASIN		Junction		HYDROBRAKE	1350	Manhole	DSG Manholes
1.001	HYDROBRAKE	1350	Manhole	DSG Manholes	OUTFALL		Junction	

Manhole Schedule

Node	Easting (m)	Northing (m)	CL (m)	Depth (m)	Dia (mm)	Connections	Link	IL (m)	Dia (mm)
BASIN	619128.497	216856.960	6.000	1.200					
						0	1.000	4.800	375
HYDROBRAKE	619133.792	216859.807	6.000	1.200	1350		1	1.000	4.800
						0	1.001	4.800	150
OUTFALL	619140.312	216856.785	5.600	0.855			1	1.001	4.745
						1			150

Simulation Settings

Rainfall Methodology	FEH-22	Summer CV	0.750	Analysis Speed	Normal	Drain Down Time (mins)	1440	Starting Level (m)	Check Discharge Volume	✓	
Rainfall Events	Singular	Winter CV	0.840	Skip Steady State	x	Additional Storage (m³/ha)	20.0	Check Discharge Rate(s)	x	100 year 360 minute (m³)	188
Storm Durations											
15 30 60 120 180 240 360 480 600 720 960 1440											
Return Period (years)	Climate Change (CC %)	Additional Area (A %)	Additional Flow (Q %)	Return Period (years)	Climate Change (CC %)	Additional Area (A %)	Additional Flow (Q %)	Return Period (years)	Climate Change (CC %)	Additional Area (A %)	Additional Flow (Q %)
2	0	0	0	30	0	0	0	100	45	4	0

Pre-development Discharge Volume

Site Makeup	Greenfield	Positively Drained Area (ha)	0.951	SPR	0.40	Return Period (years)	100	Storm Duration (mins)	360	PR	0.330
Greenfield Method	FSR/FEH	Soil Index	3	CWI	82.519	Climate Change (%)	0	Betterment (%)	0	Runoff Volume (m³)	188

Node HYDROBRAKE Online Hydro-Brake® Control

Flap Valve	x	Design Depth (m)	0.865	Sump Available	✓	Min Node Diameter (mm)	1200
Replaces Downstream Link	✓	Design Flow (l/s)	1.8	Product Number	CTL-SHE-0066-1800-0865-1800		
Invert Level (m)	4.800	Objective	(HE) Minimise upstream storage	Min Outlet Diameter (m)	0.100		

Node BASIN Depth/Area Storage Structure

Base Inf Coefficient (m/hr)	0.00000	Side Inf Coefficient (m/hr)	0.00000	Safety Factor	2.0	Porosity	1.00	Invert Level (m)	4.800	Time to half empty (mins)	
Depth (m)	Area (m²)	Inf Area (m²)	Depth (m)	Area (m²)	Inf Area (m²)						
0.000	392.0	392.0	1.200	777.0	790.1						

Results for 2 year Critical Storm Duration. Lowest mass balance: 99.99%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
480 minute winter	BASIN	456	5.028	0.228	11.7	99.9849	0.0000	OK
480 minute winter	HYDROBRAKE	456	5.028	0.228	1.8	0.3260	0.0000	SURCHARGED
15 minute summer	OUTFALL	1	4.745	0.000	1.3	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
480 minute winter	BASIN	1.000	HYDROBRAKE	1.8	0.096	0.016	0.4211	
480 minute winter	HYDROBRAKE	Hydro-Brake®	OUTFALL	1.8				125.7

Results for 30 year Critical Storm Duration. Lowest mass balance: 99.99%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
480 minute winter	BASIN	472	5.281	0.481	23.6	230.4149	0.0000	SURCHARGED
480 minute winter	HYDROBRAKE	472	5.281	0.481	1.9	0.6877	0.0000	SURCHARGED
15 minute summer	OUTFALL	1	4.745	0.000	1.8	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
480 minute winter	BASIN	1.000	HYDROBRAKE	1.9	0.096	0.017	0.6631	
480 minute winter	HYDROBRAKE	Hydro-Brake®	OUTFALL	1.8				190.3

Results for 100 year +45% CC +4% A Critical Storm Duration. Lowest mass balance: 99.99%

Node Event	US Node	Peak (mins)	Level (m)	Depth (m)	Inflow (l/s)	Node Vol (m³)	Flood (m³)	Status
960 minute winter	BASIN	945	5.660	0.860	24.1	464.8715	0.0000	SURCHARGED
960 minute winter	HYDROBRAKE	945	5.660	0.860	1.9	1.2303	0.0000	SURCHARGED
15 minute summer	OUTFALL	1	4.745	0.000	1.8	0.0000	0.0000	OK

Link Event (Upstream Depth)	US Node	Link	DS Node	Outflow (l/s)	Velocity (m/s)	Flow/Cap	Link Vol (m³)	Discharge Vol (m³)
960 minute winter	BASIN	1.000	HYDROBRAKE	1.9	0.095	0.017	0.6631	
960 minute winter	HYDROBRAKE	Hydro-Brake®	OUTFALL	1.8				230.0

12. APPENDIX E –GEOTECHNICAL REPORT



Project		KENNEDY WAY, CLACTON, ESSEX		Client		[REDACTED]		Boring Methods		LIGHT CABLE PERCUSSION 150 mm DIAMETER CASED 150 mm DIAMETER G.L. TO 1.50 m UNCASED 1.50 TO 10.00 m		Hole No.		BH1			
Ground Level		6.75 m A.O.D.		Engineer		m.E.		m.N.				Sheet		1 of 1			
Job No		8753C															
WATER				STRATA				SAMPLING/IN SITU TEST		LAB TESTING		OTHER TESTS AND NOTES					
Date/Time at Depth	Depth of Casing m	Depth to Water m	Inst.	Description	Legend	Level m.O.D.	Depth m	Depth m	Type & No.	Blows/Strength	% <425	W %	W _p %	W _L %	ρ _s Mg/m ³	C _u kN/m ²	
				Topsoil		6.65	0.10	0.10-0.60	B1								
				Made Ground (Intermixed soft dark grey brown sandy silty clay and dark grey brown silty fine-medium 'humic' sand with some subangular fine-medium flint gravel, occasional brick, concrete, ash fragments and partially decayed rootlets)		6.15	0.60	0.70	D1								pH and Water Soluble Sulphate
				Firm mottled blue grey, grey brown and orange brown slightly sandy silty CLAY with some pockets of silt and fine sand, occasional subangular-subrounded fine-medium flint gravel and rare live and partially decayed roots and rootlets (Head Deposits)		5.35	1.40	1.00-1.45	S1	N=8	84	21	21	37			
								1.45	D2								
								1.60	D3								
								2.00-2.45	U1	(30)		34			1.89	53	One-dimensional Consolidation
				Firm closely fissured mottled grey and brown slightly sandy silty CLAY with occasional blue gleying and partings of silt and fine sand (Weathered London Clay)				2.45	D4								
								2.80	D5		100	35	27	84			pH and Water Soluble Sulphate
								3.10-3.55	S2	N=12							
								3.55	D6								
								3.80	D7								
								4.00-4.45	U2	(45)		35			1.87	52	
								4.45	D8								
								4.80	D9								
								5.10-5.55	S3	N=15							
				Firm closely fissured dark grey silty CLAY with occasional partings of silt and fine sand (London Clay)		1.35	5.40	5.55	D10								pH and Water Soluble Sulphate
								5.70	D11								
								6.40-6.85	U3	(55)		34			1.92	69	
								6.85	D12								
								7.50	D13								
				- becoming firm-stiff with increasing depth				8.00-8.45	S4	N=16							
								8.45	D14								
								9.00	D15								
								9.50-9.95	U4	(60)		35			1.93	89	No groundwater recorded during drilling
								9.95	D16								Borehole complete at 10.00 m
19/05/03	1.50	DRY C						-3.25	10.00								
Water Level observations during boring, depths below GL.				WATER		SAMPLE KEY		TEST KEY		BLOWS / STRENGTH		Fieldwork		By		Mw	
Strike	Depth Obs.	5 min	10 min	15 min	20 min	1 First Strike	D Small disturbed sample	S Standard penetration test	N = N value	N = N value		19/05/03		Sheet 1 of 1		BHL	
						2 Subsequent Strike	B Bulk disturbed sample	C Cone penetration test	26/150 blows, for 150mm, drive after seating								
						N - Overnight Depth	W Water sample	K Permeability test	26*, blows for part or whole of seating drive only								
						C - Completion Depth	U Undisturbed sample	V In situ vane test	(26) U sample blow count								
						S Seepage not rising	P Piston sample		V = Vane Strength - kN/m ²								



Project		KENNEDY WAY, CLACTON, ESSEX		Client		[REDACTED]		Boring Methods		LIGHT CABLE PERCUSSION 150 mm DIAMETER CASED 150 mm DIAMETER G.L. TO 2.50 m UNCASED 2.50 TO 10.00 m		Hole No.		BH2			
Ground Level		6.77 m A.O.D.		Coordinates		m.E.		m.N.				Sheet		1 of 1			
				Engineer								Job No		8753C			
WATER				STRATA				SAMPLING/IN SITU TEST		LAB TESTING				OTHER TESTS AND NOTES			
Date/Time at Depth	Depth of Casing m	Depth to Water m	Inst.	Description	Legend	Level m.O.D.	Depth m	Depth m	Type & No.	Blows/Strength	% <425	W %	W _p %	W _L %	ρ _s Mg/m ³	C _u kN/m ²	
				Topsoil		6.67	0.10	0.20-0.70	B1								Hand excavated from ground level to 1.00 m (60 mins)
				Made Ground (Dark grey silty angular-subangular fine-medium gravel sized material composed of ash, clinker and brick fragments within a matrix of grey brown clayey fine-medium 'humic' sand)		6.47	0.30	0.80	D1								
				Medium dense grey brown slightly clayey silty very sandy subangular-subrounded fine-coarse [claybound] flint GRAVEL (Valley Gravels)				1.00-1.45	C1	N=16							Particle Size Distribution
								1.00-1.45	B2								
				Firm closely fissured mottled grey and brown slightly sandy silty CLAY with occasional blue gleying and partings of silt and fine sand (Weathered London Clay)		5.07	1.70	1.80	D2	N=8							
								2.00-2.45	S2								
								2.45	D3								
								2.80	D4								
								3.00-3.45	U1	(45)	35				1.86	52	
								3.45	D5								
								3.80	D6								
								4.10-1.55	S3	10*							
								4.55	D7								
								4.70	D8								
								5.00-5.45	U2	(40)	36				1.86	67	
								5.45	D9								
				Firm closely fissured dark grey silty CLAY with occasional partings of silt and fine sand (London Clay)		0.97	5.80	5.90	D10								
								6.50-6.95	S4	N=13							
								6.95	D11								
								7.50	D12		100	32	28	86			pH and Water Soluble Sulphate
								8.00-8.45	U3	(85)	34				1.89	62	
								8.45	D13								
								9.00	D14								pH and Water Soluble Sulphate
								9.50-9.95	S5	N=19							No groundwater recorded during drilling
								9.95	D15								Borehole complete at 10.00 m
20/05/03	2.50	DRY C				-3.23	10.00										
Water Level observations during boring, depths below GL.				WATER		SAMPLE KEY		TEST KEY		BLOWS / STRENGTH		Fieldwork		By		Sheet 1 of 1	
Strike				▼ 1 First Strike ▼ 2 Subsequent Strike N - Overnight Depth C - Completion Depth S Seepage not rising		D Small disturbed sample B Bulk disturbed sample W Water sample U Undisturbed sample P Piston sample		S Standard penetration test C Cone penetration test K Permeability test V In situ vane test		N = N value 26/150 blows, for 150mm, drive after seating 26*, blows for part or whole of seating drive only (26) U sample blow count V = Vane Strength - kN/m ²		Mw		20/05/03		BH2	
												Dates		Log		AJS	